

EXPLORATORY DATA ANALYSIS & DATA VISUALIZATION

Pima Indians Diabetes Dataset

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1. Executive Summary

This report presents the findings of an Exploratory Data Analysis (EDA) and Data Visualization project conducted on the Pima Indians Diabetes Dataset. The dataset contains health records for 768 female patients of Pima Indian heritage, all aged 21 and above, with the objective of identifying key risk factors associated with diabetes.

Through rigorous data cleaning, statistical analysis, and five comprehensive visualizations, this project uncovers the most significant predictors of diabetes and draws meaningful insights about the relationship between lifestyle and health metrics and diabetes outcomes. Glucose level emerged as the strongest single predictor, followed by BMI and Age.

2. Project Overview

2.1 Objectives

- Perform thorough exploratory data analysis on a real-world healthcare dataset
- Identify and handle data quality issues including hidden missing values
- Uncover patterns and relationships between health metrics and diabetes outcomes
- Create compelling visualizations that communicate findings clearly
- Draw actionable insights that could inform diabetes prevention strategies

2.2 Dataset

The Pima Indians Diabetes Database is one of the most widely used datasets in healthcare analytics and machine learning. It was originally collected by the National Institute of Diabetes and Digestive and Kidney Diseases and is available publicly on Kaggle via UCI Machine Learning.

Attribute	Detail
Dataset Name	Pima Indians Diabetes Database
Source	Kaggle — UCI Machine Learning
Total Records	768 patients
Number of Features	9 columns
Target Variable	Outcome (1 = Diabetic, 0 = Non-Diabetic)
Population	Female patients, Pima Indian heritage, aged 21+

Attribute	Detail
Missing Values (original)	Zero — but 616 hidden zeros found

2.3 Tools & Technologies

- Python 3 — core programming language
- Pandas & NumPy — data loading, cleaning, and statistical analysis
- Matplotlib — chart creation and customisation
- Seaborn — advanced statistical visualisations
- Jupyter Notebook — development and documentation environment

3. Methodology

3.1 Data Loading & Initial Inspection

The dataset was loaded using Pandas and immediately inspected for shape, column names, data types, and surface-level missing values. The initial inspection showed 768 rows, 9 columns, all numerical data types, and zero null values. However, further investigation revealed a critical hidden data quality issue.

3.2 Handling Hidden Missing Values

Several columns contained zero values that are medically impossible in a living patient. A blood pressure of zero, a glucose level of zero, or a BMI of zero cannot exist in reality. These zeros were identified as placeholder values used to represent missing data at the time of collection. The following columns were affected:

Column	Zero Count	% of Dataset	Action Taken
Insulin	374	48.7%	Replaced with median
SkinThickness	227	29.6%	Replaced with median
BloodPressure	35	4.6%	Replaced with median
BMI	11	1.4%	Replaced with median
Glucose	5	0.7%	Replaced with median

The median was chosen over the mean as the replacement value because columns like Insulin contained extreme outliers (maximum value of 846 against a mean of 79.8) that would significantly skew a mean-based imputation. The median is resistant to such outlier effects and provides a more representative central value.

3.3 Statistical Summary

After cleaning, descriptive statistics were generated for all columns. Key observations from the statistical summary included an average patient age of 33.2 years, an average BMI of 32.0 (technically in the obese range), an average glucose level of 120.9 (borderline elevated), and a maximum pregnancy count of 17.

4. Visualizations & Findings

Chart 1 — Diabetes Outcome Distribution

A count bar chart showing the split between diabetic and non-diabetic patients. Of 768 patients, 500 (65.1%) were non-diabetic, and 268 (34.9%) were diabetic. This confirms a meaningful class imbalance that would need to be addressed if building a predictive model.

Chart 2 — Distribution of Health Metrics by Outcome

A 3x3 grid of histogram plots showing the distribution of each health metric, split by diabetes outcome. This chart immediately reveals which metrics show the clearest separation between diabetic and non-diabetic patients. Glucose showed the most dramatic separation, with diabetic patients clustering heavily toward higher values. Age and BMI also showed clear rightward shifts for diabetic patients, while BloodPressure distributions overlapped heavily between both groups.

Chart 3 — Correlation Heatmap

A heatmap displaying the correlation coefficient between every pair of variables in the dataset. Focusing on the Outcome row, Glucose had the strongest correlation at 0.49, followed by BMI at 0.31, Age at 0.24, and Pregnancies at 0.22. BloodPressure had the weakest correlation at 0.17. A notable secondary finding was the strong correlation between SkinThickness and BMI (0.54), which is biologically expected as higher body fat produces both higher BMI and thicker skin folds.

Chart 4 — Box Plots by Outcome

Side-by-side box plots comparing the distribution of each health metric between diabetic and non-diabetic patients. The Glucose box plot showed the most dramatic separation; the entire diabetic box sat above the non-diabetic box with minimal overlap. The Insulin box plots revealed numerous extreme outliers in both groups, reflecting the high variability in insulin levels across the patient population.

Chart 5 — Pair Plot

A comprehensive pairwise scatter plot matrix showing the relationship between every combination of health metrics, coloured by diabetes outcome. The Glucose vs Age and Glucose vs BMI scatter plots were particularly revealing, showing diabetic patients (pink) clustering distinctly in the high-glucose, high-age and high-glucose, high-BMI corners respectively. This multi-dimensional view confirmed that diabetes risk is highest when multiple risk factors are elevated simultaneously.

5. Key Findings

Finding	Detail
Diabetes prevalence	34.9% of patients in the dataset are diabetic
Strongest predictor	Glucose (correlation: 0.49)
Average glucose gap	Diabetic: 142.1 vs Non-Diabetic: 110.7 (31.4 point difference)
Second strongest predictor	BMI (correlation: 0.31)
Average BMI gap	Diabetic: 35.4 vs Non-Diabetic: 30.9
Age difference	Diabetic average: 37.1 years vs Non-Diabetic: 31.2 years
Weakest predictor	BloodPressure (correlation: 0.17)
Data quality issue	616 medically impossible zeros found and replaced across 5 columns

6. Recommendations

Recommendation 1: Prioritise glucose monitoring as the primary screening tool

Given that glucose level is by far the strongest predictor of diabetes in this dataset, regular blood glucose testing should be the cornerstone of any diabetes screening programme. The 31.4 point average difference between diabetic and non-diabetic patients makes glucose a highly reliable early warning indicator.

Recommendation 2: Address the obesity-diabetes link urgently

With BMI being the second strongest predictor and both groups already averaging in the overweight-to-obese range (30.9 and 35.4 respectively), weight management

interventions could have a significant impact on diabetes prevention. Public health programmes targeting BMI reduction in at-risk populations are strongly indicated.

Recommendation 3: Implement age-targeted screening programmes

The 6-year average age difference between diabetic and non-diabetic patients suggests that diabetes risk accelerates meaningfully with age. Routine diabetes screening should be introduced as a standard component of healthcare for patients from age 35 onwards, with increased frequency for those with additional risk factors.

Recommendation 4: Improve data collection practices

The discovery of 616 medically impossible zero values, particularly the 374 zero insulin readings representing nearly 49% of all insulin data, highlights a serious data collection issue in this dataset. Healthcare data systems should implement validation rules that prevent physiologically impossible values from being recorded, ensuring data quality at the point of entry rather than requiring correction during analysis.

Recommendation 5: Consider multiple risk factors together

The pair plot analysis revealed that diabetes risk is highest when multiple factors are elevated simultaneously, particularly high glucose combined with high BMI or advanced age. Clinical risk assessment tools should incorporate multi-factor scoring rather than relying on any single metric in isolation.

7. Conclusion

This project successfully demonstrated the power of Exploratory Data Analysis and Data Visualization to extract meaningful insights from a real-world healthcare dataset. Through careful data cleaning, statistical analysis, and five comprehensive visualizations, the analysis identified glucose level, BMI, and age as the three most significant risk factors for diabetes in this patient population.

Beyond the technical exercise, the findings carry genuine public health relevance. In a world where diabetes affects over 500 million people globally and continues to rise, particularly in developing nations, the ability to identify and communicate risk factors clearly through data is a genuinely valuable contribution. Early detection through glucose monitoring, weight management, and age-targeted screening could prevent countless cases before they develop into full diagnoses.

This analysis forms a strong foundation for future predictive modelling work, where the insights gained here about feature importance and data quality could directly inform a more accurate and reliable diabetes prediction system.

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